Feature Importance for Tabular Predictive Models

Post-hoc Model Agnostic Methods Based on Cooperative Game Theory

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Bootcamp: Statistical Insights into Modern AI Systems
IVADO Thematic Semester - Statistical Foundations of AI
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Black-box model interpretability
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Research interests:
Statistical Learning • XAI • Uncertainty Quantification • Sensitivity Analysis • Probability Theory • Cooperative game theory • Functional analysis

Agenda

Day 4 - May 4, 2026

9:30 AM – 11:00 AM – Lecture 1: Feature importance and Cooperative Games ← We are here

11:00 AM – 11:30 AM – Coffee Break

11:30 AM – 12:15 PM – Lab 1: Cooperative Games from scratch (Python)

12:15 PM – 2:00 PM – Lunch

2:00 PM – 2:45 PM – Lab 2: Showcase of various use-cases (Python & R)

2:45 PM – 3:30 PM – Lecture 2: Exploring concepts in LLMs and Diffusion models

3:30 PM – 4:00 PM – Coffee Break

4:00 PM – 4:45 PM – Lab 3: SAEs for LLMs and Diffusion Models

Predictive modeling: different mindsets

We can approach predictive modeling from different perspectives (Molnar 2023):

- Statistical mindset: predictive models are **tools for studying multivariate associations**
  ☞ Emphasis on **understanding** the phenomena
- Machine learning (ML) mindset: predictive models are **tools for accomplishing tasks**
  ☞ Emphasis on **mimicking** the phenomena

The ML mindset led the way to tackle highly non-trivial tasks
☞ e.g., image (Krizhevsky et al. 2017) and text (Yang et al. 2019) classification

These advances came at the cost of **an increased model complexity**:

We can show that these predictive model work...
... but we don't know why

When AI goes wrong... and right!

Failures	Successes
A US court algorithm labeled Black defendants as future criminals twice as often as white ones. Used in real sentencing decisions, with no way to challenge it.	**An AI predicted the 3D shape of nearly every known protein on Earth**, about 200 million of them. Its creators won the 2024 Nobel Prize in Chemistry.
During COVID, AI models claimed astounding accuracy at diagnosing the disease from chest scans. They were actually reading hospital markings, not lungs. Useless on real patients.	**An AI screens diabetic patients for blindness in countries with too few eye doctors.** Over 600,000 people screened so far in India and Thailand, on par with specialists.
AI built to rank job candidates. It systematically downgraded women. It had learned from a decade of male-dominated hiring.	**An AI is on-par with the world's best weather supercomputers on 10-day forecasts**. It runs in minutes instead of hours with remarkable accuracy.

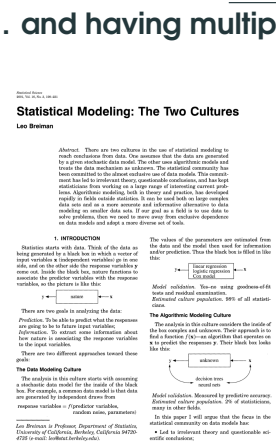
Being able to understand these predictive models would have helped avoid failures

My (nuanced) take

Modeling mindsets feed off each other:

There is not a right way to approach predictive modeling...

... and having multiple perspectives can only benefit the analysis



Statistical Science
2001, Vol. 16, No. 3, 199–231

Statistical Modeling: The Two Cultures

Leo Breiman

Abstract. There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. The statistical community has been committed to the almost exclusive use of data models. This commitment has led to irrelevant theory, questionable conclusions, and has kept statisticians from working on a large range of interesting current problems. Algorithmic modeling, both in theory and practice, has developed rapidly in fields outside statistics. It can be used both on large complex data sets and as a more accurate and informative alternative to data modeling on smaller data sets. If our goal as a field is to use data to solve problems, then we need to move away from exclusive dependence on data models and adopt a more diverse set of tools.

1. INTRODUCTION

Statistics starts with data. Think of the data as being generated by a black box in which a vector of input variables **x** (independent variables) go in one side, and on the other side the response variables **y** come out. Inside the black box, nature functions to associate the predictor variables with the response variables, so the picture is like this:

There are two goals in analyzing the data:

Prediction. To be able to predict what the responses are going to be to future input variables;

Information. To extract some information about how nature is associating the response variables to the input variables.

There are two different approaches toward these goals:

The Data Modeling Culture

The analysis in this culture starts with assuming a stochastic data model for the inside of the black box. For example, a common data model is that data are generated by independent draws from

response variables = f(predictor variables, random noise, parameters)

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198

The values of the parameters are estimated from the data and the model then used for information and/or prediction. Thus the black box is filled in like this:

Model validation. Yes–no using goodness-of-fit tests and residual examination.

Estimated culture population. 98% of all statisticians.

The Algorithmic Modeling Culture

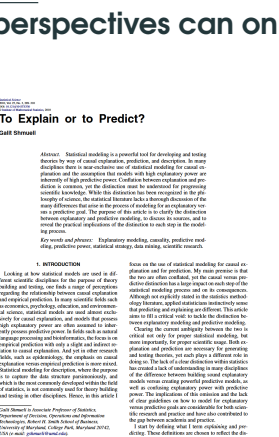
The analysis in this culture considers the inside of the box complex and unknown. Their approach is to find a function $f(\mathbf{x})$—an algorithm that operates on **x** to predict the responses **y**. Their black box looks like this:

Model validation. Measured by predictive accuracy.

Estimated culture population. 2% of statisticians, many in other fields.

In this paper I will argue that the focus in the statistical community on data models has:

- Led to irrelevant theory and questionable scientific conclusions;



Statistical Science
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To Explain or to Predict?

Galit Shmueli

Abstract. Statistical modeling is a powerful tool for developing and testing theories by way of causal explanation, prediction, and description. In many disciplines there is near-exclusive use of statistical modeling for causal explanation and the assumption that models with high explanatory power are inherently of high predictive power. Confusion between explanation and prediction is common, yet the distinction must be understood for progressing scientific knowledge. While this distinction has been recognized in the philosophy of science, the statistical literature lacks a thorough discussion of the many differences that arise in the process of modeling for an explanatory versus a predictive goal. The purpose of this article is to clarify the distinction between explanatory and predictive modeling, to discuss its sources, and to reveal the practical implications of the distinction to each step in the modeling process.

Key words and phrases: Explanatory modeling, causality, predictive modeling, predictive power, statistical strategy, data mining, scientific research.

1. INTRODUCTION

Looking at how statistical models are used in different scientific disciplines for the purpose of theory building and testing, one finds a range of perceptions regarding the relationship between causal explanation and empirical prediction. In many scientific fields such as economics, psychology, education, and environmental sciences, statistical models are used almost exclusively for causal explanation, and a model attaining high explanatory power are often assumed to inherently possess predictive power. In fields such as natural language processing and bioinformatics, the focus is on empirical prediction with only a slight and indirect relation to causal explanation. And yet in other research fields, such as epidemiology, the emphasis on causal explanation versus empirical prediction is more mixed. Statistical modeling for description, where the purpose is to capture the data structure parsimoniously, and which is the most commonly developed within the field of statistics, is not commonly used for theory building and testing in other disciplines. Hence, in this article I focus on the use of statistical modeling for causal explanation and for prediction. My main premise is that to ignite a productive statistical research agenda, a clear distinction has a large impact on each step of the statistical modeling process and on its consequences. Although not explicitly stated in the statistics methodology literature, applied statisticians instinctively sense that predicting and explaining are different. This article aims to fill a critical void: to tackle the distinction between explanatory modeling and predictive modeling.

Clearing the current ambiguity between the two is critical not only for proper statistical modeling, but more importantly, for proper scientific usage. Both explanation and prediction are necessary for generating and testing theories, yet each plays a different role in doing so. The lack of a clear distinction within statistics has created a lack of understanding in many disciplines of how statistical modeling can be misused when causal models versus creating powerful predictive models, as well as confusing explanatory power with predictive power. The implications of this omission and the lack of clear guidelines on how to model for explanatory versus predictive goals are considerable for both scientific research and practice and have also contributed to the gap between academia and practice.

I start by defining what I term *explaining* and *predicting*. These definitions are chosen to reflect the dis-

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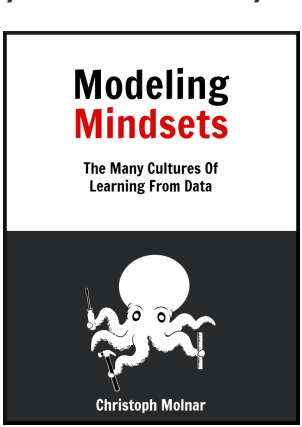
289



Goals for this lecture

Spirit of this lecture: somewhere **in-between** the two mindsets

By lunch, I want **you** to:

- Know **why** explainable artificial intelligence is important
- Understand **how** attribution methods based on cooperative games work
- Have an idea of **why** they work
- Discuss their **limitations** and **challenges**
- Be **confident enough** to implement these types of methods from scratch

I need your help to meet these goals:

Please, feel free to interrupt at any point to ask questions

(You can ask them in french or in english)

Outline

1. Context and preliminaries
2. Cooperative games for feature importance
3. In practice
4. Discussion

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Decision-making in high-stakes applications

Why are we (rightfully) apprehensive about using AI for critical systems?

Critical system: A system whose failure leads to **dramatic consequences**
e.g. hydroelectric dam, aircraft, financial market, judicial system, a human body

Main reason: Decision-making must be **justifiable** and **justified**
e.g. using statistical arguments

This one of the goals that **eXplainable Artificial Intelligence (XAI)** aims to solve

The many goals of XAI

The field of XAI is **very broad**

☞ It encompasses every aspects of explaining algorithms

- **Taxonomy of models** (white-box, glass-box, grey-box, black-box...)
- **Human-AI interactions** (user interfaces, design...)
- **Interpretability/explainability of models** (methods, evaluation, benchmarks...)
- **Philosophical aspects** (formal logic, definition of explaining...)

Settings that XAI aim to address (Barredo Arrieta et al. 2020):

Trustworthiness • Causality • Transferability • Informativeness • Confidence • Fairness • Accessibility • Interactivity • Privacy awareness

A lot of people wrote a lot of ideas with very different perspectives

Which can sometimes be a little confusing...

Framework

This lecture's focus: **post-hoc model-agnostic XAI methods** based on **cooperative games**

- Post-hoc: the model is **already fitted**
- Model-agnostic: no reliance on a **model architecture** (e.g., neural net, random forest...)
- XAI methods: procedures to **extract insights** from a predictive model
- Cooperative games: **attribute "importance"** to the covariates

Remark.

☞ Black-box model $\neq$ complex model
It is the **expression of model-agnosticity**

☞ Post-hoc approach
The model is fixed, but we can make predictions with it

Motivating examples

Example.

You are an auditor for Québec's Autorité des Marchés Financiers (financial markets regulator).

You needs to assess if an insurance company's pricing model discriminates against protected groups.

You do not know how the model has been built. For auditing purposes, you can query the model and get predictions back.

How do you decide whether the predictive model is suitable or not?

Example.

You are a radiologist at the Hotel-Dieu de Québec hospital using predictive models to detect early-stage lung cancer.

The model identifies a high-risk in a young patient with a healthy lifestyle.

The software is a proprietary black-box system. You must decide whether to send the patient for an very invasive biopsy based on this prediction.

How do you decide whether to perform the surgical intervention or not?

Notations

☞ $X = (X_1, \ldots, X_d)$ is a random vector of d covariates, valued in $\mathbb{R}^d$

☞ $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$ denotes an **observation** of x

☞ $\widehat{f} : \mathbb{R}^d \to \mathbb{R}$ is the predictive model

☞ $\widehat{f}(X)$ is a random variable called **the random output**

☞ $\widehat{f}(x) \in \mathbb{R}$ is the prediction for the observation x

☞ $D = \{1, \ldots, d\}$ is the set of indices of the covariates

☞ $\mathcal{P}_D$ is the **set of subsets of** D

If $d = 3$ then $D = \{1, 2, 3\}$, $\mathcal{P}_D = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$ and $|\mathcal{P}_D| = 2^d$

☞ For $A \in \mathcal{P}_D$, X_A is the **subset of covariables whose indices are in** A

If $A = \{1, 3\}$, $X_A = (X_1, X_3)$. By convention $X_\emptyset \in \mathbb{R}$ and $X_D = X$

Outline

1. Context and preliminaries

2. Cooperative games for feature importance

3. In practice

4. Discussion

Cooperative games

"Cooperative games = The art of sharing a cake"

Two ingredients:

☞ A set of players: $D = \{1, \ldots, d\}$

$\mathcal{P}_D$ represents the set of **coalitions** (i.e., teams)

☞ A **value function**: $v : \mathcal{P}_D \to \mathbb{R}$

It assigns a **value** to each **coalition**

(We assume WLOG that $v(\emptyset) = 0$)

(D, v) formally defines **a cooperative game**

The goal:

☞ **Redistribute $v(D)$ (the cake) among the d players**

Cooperative games in the wild - LMG indices

Lindeman et al. (1980) : Measure the covariates' importance in a linear regression model

Their idea: Redistribute the full model's R^2 back to the covariates

☞ Players: The covariates $X_1, \ldots, X_d$

☞ Coalitions: The subsets of covariates

☞ **Value function:** $v(A) = R^2_Y(X_A)$, the R^2 of the **nested model fit by only using** X_A

☞ The cake: The R^2 coefficient of the **full model** $v(D) = R^2_Y(X)$

Computing the **value function** for **every nested models** $\implies 2^d$ coefficients

For 20 covariates: more than **a million** numbers

☞ **How can we process all of this information?**

Allocations

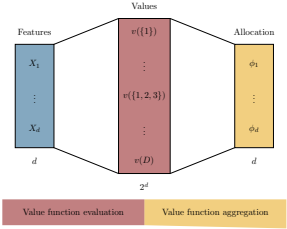
Allocation : **Aggregates** the evaluations of the **value function**

Summarizes the 2^d numbers into **one quantity for each of the players**

It is a mapping $\phi : D \to \mathbb{R}$, that must respect the **efficiency criteria**: $\sum_{i \in D} \phi(i) = v(D)$

Ensures that **we actually redistribute the cake**

In a nutshell:



☞ We want to study a model with d covariates

☞ We compute **the value function** for each of the 2^d subsets of covariates

☞ We aggregate the 2^d numbers into d numbers using an **efficient allocation**

☞ **How do we define efficient allocations?**

Shapley values

In the cooperative game theory literature, **allocations** are characterized using **axioms**

Shapley (1951) values: **unique** allocation $\mathrm{Shap} : D \to \mathbb{R}$ that respects

- **Efficiency:** $\sum_{i=1}^{d} \mathrm{Shap}(i) = v(D)$ ✔
- **Symmetry/Anonymity:** if, for every $A \in \mathcal{P}_{D \setminus \{i,j\}}$ $v(A \cup \{i\}) = v(A \cup \{j\})$, then $\mathrm{Shap}(i) = \mathrm{Shap}(j)$ ✗
- **Dummy player:** if, for every $A \in \mathcal{P}_D$, $v(A \cup \{i\}) = v(A)$, then $\mathrm{Shap}(i) = 0$ ✗
- **Additivity:** The sum of Shap from games (D, v) and (D, w) is equal to the Shap of the game $(D, v + w)$ ✗

For every $i \in D$, it writes:

$$\mathrm{Shap}(i) = \sum_{A \in \mathcal{P}_D \,:\, i \notin A} \frac{|A|!\,(d - |A| - 1)!}{d!} \left[v\left(A \cup \{i\}\right) - v(A) \right]$$

Major result which revolutionized cooperative game theory...

... but it is not **the only solution** to **every** XAI settings

☞ **Can we go beyond the Shapley values?**

Allocations as a dividend sharing mechanism

The **Harsanyi (1963) dividends** are defined as:

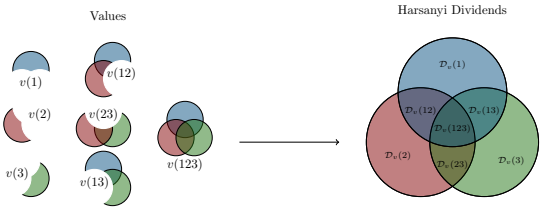
$$\mathcal{D}_v(A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} v(B), \quad \text{or equivalently,} \quad \mathcal{D}_v(A) = v(A) - \sum_{B \in \mathcal{P}_A} \mathcal{D}_v(B)$$

Measure the **surplus/added-value/synergy of a coalition**:

$$\mathcal{D}_v(12) = v(12) - v(1) - v(2)$$

And they also respect a **telescopic sum property**:

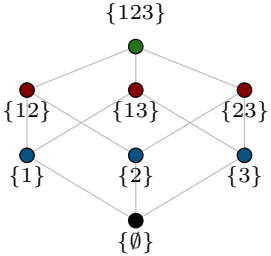
$$\forall A \in \mathcal{P}_D, \quad v(A) = \sum_{B \in \mathcal{P}_A} \mathcal{D}_v(B)$$



Allocations as a dividend sharing mechanism

Where does this telescopic sum property come from?

A	$v(123)$	$v(12)$	$v(23)$	$v(13)$	$v(3)$	$v(2)$	$v(1)$
$\mathcal{D}_v(123)$	+1	-1	-1	-1	+1	+1	+1
$\mathcal{D}_v(12)$		+1			-1	-1	
$\mathcal{D}_v(23)$			+1			-1	-1
$\mathcal{D}_v(13)$				+1	-1		-1
$\mathcal{D}_v(1)$					+1		
$\mathcal{D}_v(2)$						+1	
$\mathcal{D}_v(3)$							+1
Sum	**1**	0	0	0	0	0	0



$\mathcal{P}_{\{1,2,3\}}$ has a Boolean lattice structure

Proposition *(Möbius inversion on the Boolean lattice (Rota 1964; Kung et al. 2012))*

For any two set functions $v : \mathcal{P}_D \to \mathbb{R}$, $\mathcal{D} : \mathcal{P}_D \to \mathbb{R}$, the following equivalence holds:

$$\forall A \in \mathcal{P}_D, \quad v(A) = \sum_{B \in \mathcal{P}_A} \mathcal{D}(B), \quad \iff \quad \forall A \in \mathcal{P}_D, \quad \mathcal{D}(A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} v(B).$$

Allocations as a dividend sharing mechanism

The **Harsanyi set of allocations** is a **family of efficient allocations**

They are defined as an **aggregation of dividends**

$$\phi(i) = \sum_{A \in \mathcal{P}_D \,:\, i \in A} \lambda_i(A) \mathcal{D}_v(A), \quad \text{where} \quad \begin{cases} \forall i \in D,\ \forall A \in \mathcal{P}_D,\ \ \lambda_i(A) \geq 0, \\ \forall A \in \mathcal{P}_D,\ \ \sum_{i \in D} \lambda_i(A) = 1 \end{cases}$$

parametrized by a **weight system** $\lambda : D \times \mathcal{P}_D \to [0, 1]$

<u>Idea:</u> the **added-value** of each coalition **is redistributed among its players**

☞ How? According to λ!

Why are these allocations efficient?

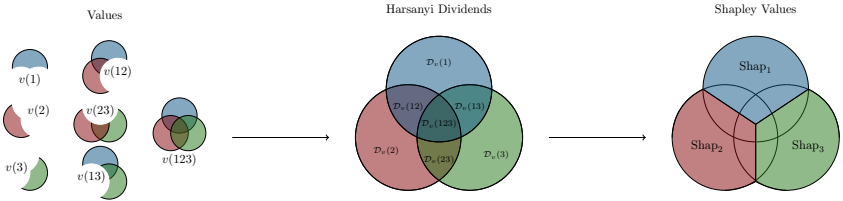
$$\sum_{i=1}^{d} \phi(i) = \sum_{i=1}^{d} \sum_{A \in \mathcal{P}_D} \lambda_i(A) \mathbb{1}_{\{i \in A\}} \mathcal{D}_v(A) = \underbrace{\left(\sum_{A \in \mathcal{P}_D} \mathcal{D}_v(A) \right)}_{=v(D)} \underbrace{\left(\sum_{i=1}^{d} \lambda_i(A) \mathbb{1}_{\{i \in A\}} \right)}_{=1} = v(D)$$

Example of allocations in the Harsanyi set

The **Shapley values**: **Egalitarian redistribution** of the **dividends**

Split the added-value equally among the players in the coalitions

$$\lambda_i(A) = \frac{1}{|A|}$$



The **proportional Shapley values**: **Proportional redistribution** of the **dividends**

Proportionality is understood w.r.t. individual contributions

$$\lambda_i(A) = \frac{|v(\{i\})|}{\sum_{j \in A} |v(\{j\})|}$$

Analytical illustrations - Shapley and conditional expectation

$$\widehat{f}(X) = X_1 + X_1 X_2, \quad \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right), \quad -1 < \rho < 1$$

Conditional expectation: $v_{\mathrm{C}}(A) = \mathbb{E}\left[\widehat{f}(X) \mid X_A = x_A\right]$ for an observation $x \in \mathbb{R}^d$

$$v_{\mathrm{C}}(\emptyset) = 0, \quad v_{\mathrm{C}}(\{1\}) = x_1(1 + x_1\rho), \quad v_{\mathrm{C}}(\{2\}) = x_2\rho(1 + x_2), \quad \boxed{v_{\mathrm{C}}(\{12\}) = x_1 + x_1 x_2 = \widehat{f}(x)}$$

$$\mathcal{D}_{v_{\mathrm{C}}}(\emptyset) = 0, \quad \mathcal{D}_{v_{\mathrm{C}}}(\{1\}) = x_1(1 + \rho x_1), \quad \mathcal{D}_{v_{\mathrm{C}}}(\{2\}) = \rho x_2(1 + \rho x_2)$$

$$\mathcal{D}_{v_{\mathrm{C}}}(\{1,2\}) = \underbrace{x_1 + x_1 x_2}_{v_{\mathrm{C}}(\{1,2\})} - \underbrace{x_1(1 + \rho x_1)}_{\mathcal{D}_{v_{\mathrm{C}}}(\{1\})} - \underbrace{\rho x_2(1 + \rho x_2)}_{\mathcal{D}_{v_{\mathrm{C}}}(\{2\})}$$

Shapley values with **conditional expectation**: ($\lambda_i(A) = \frac{1}{|A|}$)

$$\mathrm{Shap}(1) = 1 \times \mathcal{D}_{v_{\mathrm{C}}}(\{1\}) + \frac{1}{2} \times \mathcal{D}_{v_{\mathrm{C}}}(\{1,2\}) = \frac{1}{2}\left[x_1 + x_1 x_2 + x_1(1 + \rho x_1) - \rho x_2(1 + \rho x_2)\right]$$

$$\mathrm{Shap}(2) = 1 \times \mathcal{D}_{v_{\mathrm{C}}}(\{2\}) + \frac{1}{2} \times \mathcal{D}_{v_{\mathrm{C}}}(\{1,2\}) = \frac{1}{2}\left[x_1 + x_1 x_2 - x_1(1 + \rho x_1) + \rho x_2(1 + \rho x_2)\right]$$

This is the (infamous) `SHAP` method (Lundberg et al. 2017)

Analytical illustrations - Proportional Shapley and marginal expectation

$$\widehat{f}(X) = X_1 + X_1 X_2, \quad \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right), \quad -1 < \rho < 1$$

Marginal expectation: $v_m(A) = \mathbb{E}\left[\widehat{f}(x_A, X_{\sim A})\right] = \int_{\mathbb{R}^{|\sim A|}} f(x_A, x_{\sim A}) dP_{X_{\sim A}}(x_{\sim A})$

$$v_m(\emptyset) = 0, \quad v_m(\{1\}) = x_1, \quad v_m(\{2\}) = 0, \quad \boxed{v_m(\{12\}) = x_1 + x_1 x_2 = \widehat{f}(x)}$$

$$\mathcal{D}_{v_m}(\emptyset) = 0, \quad \mathcal{D}_{v_m}(\{1\}) = x_1, \quad \mathcal{D}_{v_m}(\{2\}) = 0$$

$$\mathcal{D}_{v_m}(\{1,2\}) = \underbrace{x_1 + x_1 x_2}_{v_m(\{1,2\})} - \underbrace{x_1}_{\mathcal{D}_{v_m}(\{1\})} - \underbrace{0}_{\mathcal{D}_{v_m}(\{2\})} = x_1 x_2$$

Proportional Shapley values with **conditional expectation**: $(\lambda_i(A) = \frac{|v(\{i\})|}{\sum_{j \in A} |v(\{j\})|})$

$$\text{P-Shap}(1) = 1 \times \mathcal{D}_{v_m}(\{1\}) + \frac{|\mathcal{D}_{v_m}(\{1\})|}{|\mathcal{D}_{v_m}(\{1\})| + |\mathcal{D}_{v_m}(\{2\})|} \times \mathcal{D}_{v_m}(\{1,2\}) = x_1 + x_1 x_2$$

$$\text{P-Shap}(2) = 1 \times \mathcal{D}_{v_m}(\{2\}) + \frac{|\mathcal{D}_{v_m}(\{2\})|}{|\mathcal{D}_{v_m}(\{1\})| + |\mathcal{D}_{v_m}(\{2\})|} \times \mathcal{D}_{v_m}(\{1,2\}) = 0$$

Allocations and random orders

The **Weber (1988) set** is a family of **efficient allocations** based on **random orders**

☞ **Dynamic approach** of the value-sharing mechanism

An **order** is represented by **a permutation** of D

☞ $\mathcal{S}_D$ is the set of permutations of D

There are $d!$ permutations in $\mathcal{S}_D$

☞ For $d = 3$, we have $\mathcal{S}_D = \{(1,2,3),(1,3,2),(2,3,1),(3,2,1),(2,1,3),(3,1,2)\}$

For an order $\pi = (\pi_1, \ldots, \pi_d) \in \mathcal{S}_D$, we denote $\pi(i)$ the position of player i in π

☞ It allows us to track where i is in π, i.e., $\pi_{\pi(i)} = i$

For an order $\pi \in \mathcal{S}_D$ and a player $i \in D$, we write $\pi^i = \{\pi_1, \ldots, \pi_{\pi(i)}\}$

☞ All the players up to i (included) in the permutation

Allocations and random orders

The **marginal contribution** of a player $i \in D$ to an order $\pi \in \mathcal{S}_D$ is

$$\mathrm{MC}^i_\pi = v(\pi^i) - v(\pi^i \setminus \{i\})$$

For the permutation $\pi = (3, 1, 2)$, we have:

$$\mathrm{MC}^1_\pi = v(\{1,3\}) - v(\{3\}), \quad \mathrm{MC}^2_\pi = v(\{1,2,3\}) - v(\{2,3\}), \quad \mathrm{MC}^3_\pi = v(\{3\})$$

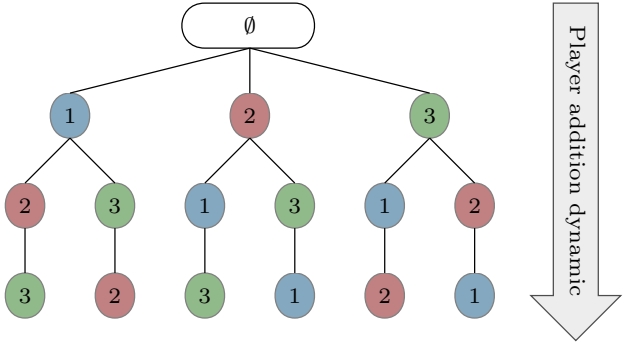
The **Weber set** is defined as the **expected marginal contribution**

$$\begin{aligned}\phi(i) &= \sum_{\pi \in \mathcal{S}_D} p(\pi) \times \mathrm{MC}^i_\pi \\ &= \mathbb{E}_{\pi \sim p}\left[v(\pi^i) - v(\pi^i \setminus \{i\})\right]\end{aligned}$$

parametrized by a **probability mass function (pmf)** $p : \mathcal{S}_D \to [0, 1]$ over $\mathcal{S}_D$

☞ Assign a probability to every permutation

Dynamic interpretation



$(1,2,3)$ $(1,3,2)$ $(2,1,3)$ $(2,3,1)$ $(3,1,2)$ $(3,2,1)$ $\mathcal{S}_D$

$$\mathrm{MC}^i_\pi = v\left(C_{\pi(i)}(\pi)\right) - v\left(C_{\pi(i)-1}(\pi)\right)$$

$$\phi_1 = \sum_{\pi \in \mathcal{S}_D} p(\pi) \times \mathrm{MC}^1_\pi$$

$$\phi_2 = \sum_{\pi \in \mathcal{S}_D} p(\pi) \times \mathrm{MC}^2_\pi$$

$$\phi_3 = \sum_{\pi \in \mathcal{S}_D} p(\pi) \times \mathrm{MC}^3_\pi$$

Allocations and random orders

There is also a telescopic sum property:

$$\sum_{i \in D} \mathrm{MC}^i_\pi = v(D)$$

Ordering	MC^1	MC^2	MC^3	Sum
$(1,2,3)$	$v(\{1\})$	$v(\{1,2\}) - v(\{1\})$	$v(\{1,2,3\}) - v(\{1,2\})$	$v(\{1,2,3\})$
$(1,3,2)$	$v(\{1\})$	$v(\{1,2,3\}) - v(\{1,3\})$	$v(\{1,3\}) - v(\{1\})$	$v(\{1,2,3\})$
$(2,3,1)$	$v(\{1,2,3\}) - v(\{2,3\})$	$v(\{2\})$	$v(\{2,3\}) - v(\{2\})$	$v(\{1,2,3\})$
$(3,2,1)$	$v(\{1,2,3\}) - v(\{2,3\})$	$v(\{2,3\}) - v(\{3\})$	$v(\{3\})$	$v(\{1,2,3\})$
$(2,1,3)$	$v(\{1,2\}) - v(\{2\})$	$v(\{2\})$	$v(\{1,2,3\}) - v(\{1,2\})$	$v(\{1,2,3\})$
$(3,1,2)$	$v(\{1,3\}) - v(\{3\})$	$v(\{1,2,3\}) - v(\{2,3\})$	$v(\{3\})$	$v(\{1,2,3\})$

(Recall we assumed $v(\emptyset) = 0$)

Why are these allocations efficient?

$$\sum_{i \in D} \phi(i) = \sum_{i \in D} \sum_{\pi \in \mathcal{S}_D} p(\pi) \times \mathrm{MC}^i_\pi = \underbrace{\sum_{\mathcal{S}_D} p(\pi)}_{=1} \underbrace{\sum_{i \in D} \mathrm{MC}^i_\pi}_{=v(D)} = v(D)$$

Example of allocations in the Weber set

The **Shapley values**: **Equi-probabilistic orders**, i.e., **uniform distribution** over $\mathcal{S}_D$

Split the added-value equaly among the players in the coalitions

$$p(\pi) = d!$$

The **proportional Shapley values** (Dehez 2017):

Proportionality is understood w.r.t. individual contributions

$$p(\pi) = \exp\left[-\sum_{j=2}^{d} \log\left(1 + \sum_{k=1}^{d-1} \frac{|v(\pi_j)|}{|v(\pi_k)|}\right)\right]$$

The **proportional values** (Ortmann 2000):

Proportionality is understood w.r.t. all contributions

$$p(\pi) = \frac{L(\pi)}{\sum_{\sigma \in \mathcal{S}_D} L(\sigma)}, \quad L(\pi) = \exp\left(-\sum_{j \in D} \log\left(v\left(\{\pi_1, \dots \pi_{\pi(j)}\}\right)\right)\right)$$

Analytical illustrations - Shapley and variance

$$\widehat{f}(X) = X_1 + X_2, \quad X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \rho \\ 0 & 1 & 0 \\ \rho & 0 & 1 \end{pmatrix} \right)$$

Variance of the conditional expectation (closed Sobol' (2001) index): $v(A) = \mathbb{V}\left(\mathbb{E}\left[\widehat{f}(X) \mid X_A\right]\right)$

$$v(\emptyset) = 0, \quad v(\{1\}) = 1, \quad v(\{2\}) = 1, \quad v(\{3\}) = \rho^2, \quad v(\{12\}) = 2, \quad v(\{13\}) = 1, \quad v(\{23\}) = 1 + \rho^2,$$

$$\boxed{v(\{123\} = 2 = \mathbb{V}\left(\widehat{f}(X)\right)}$$

Shapley values with **closed Sobol' indices**: ($p(\pi) = 1/d!$)

$$\mathrm{Shap}(1) = \underbrace{\frac{2}{6} v(\{1\})}_{(1,2,3)\ (1,3,2)} + \underbrace{\frac{2}{6}\left[v(\{123\}) - v(\{23\})\right]}_{(2,3,1)\ (3,2,1)} + \underbrace{\frac{1}{6}\left[v(\{12\}) - v(\{2\})\right]}_{(2,1,3)} + \underbrace{\frac{1}{6}\left[v(\{13\}) - v(\{3\})\right]}_{(3,1,2)} = 1 - \rho^2/2$$

$$\mathrm{Shap}(2) = 1$$

$$\mathrm{Shap}(3) = \rho^2/2$$

These are (essentially) the Shapley Effects (Owen 2014)

Analytical illustrations - Proportional values and variance

$$\widehat{f}(X) = X_1 + X_2, \quad X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \rho \\ 0 & 1 & 0 \\ \rho & 0 & 1 \end{pmatrix} \right)$$

Expectation of the conditional variance (total Sobol' index): $v(A) = \mathbb{E}\left[\mathbb{V}\left(\widehat{f}(X) \mid X_{\sim A}\right)\right]$

$$v(\emptyset) = 0, \quad v(\{1\}) = 1 - \rho^2, \quad v(\{2\}) = 1, \quad v(\{3\}) = 0, \quad v(\{12\}) = 2 - \rho^2, \quad v(\{13\}) = 1, \quad v(\{23\}) = 1,$$

$$\boxed{v(\{123\} = 2 = \mathbb{V}\left(\widehat{f}(X)\right)}$$

Proportional values with **total Sobol' indices**: (+ a little bit of work)

$$\mathrm{PV}(1) = 1$$

$$\mathrm{PV}(2) = 1$$

$$\mathrm{PV}(3) = 0$$

These are (essentially) the proportional marginal effects (PME) (Herin et al. 2024)

Relationship between the Harsanyi set and the Weber set

The **Harsanyi set** is **bigger** than the **Weber set**

In the sense that the Harsanyi set contains the Weber set

We can retrieve a **weight-system** λ (Harsanyi) from a **pmf** p **over** $\mathcal{S}_D$ (Weber) (Derks 2005)

$$\forall i \in D,\ \forall A \in \mathcal{P}_D, \quad \lambda_i(A) = \sum_{\pi \in \mathcal{S}_D\ :\ A \in \pi^i} p(\pi)$$

☞ Sum the probabilities of all the orders for which all the players up to i coincides with A

The recipe

Three fundamental steps to use **cooperative games** for decompositions (I. et al. 2025b):

- Step 1: Choose a **quantity of interest** (the cake)
  e.g., a prediction $\widehat{f}(x)$, the model's variance $\mathbb{V}\left(\widehat{f}(X)\right)$, a performance metric $R^2_Y(X)$
  ☞ Guides **the overall interpretation** of the method

- Step 2: Choose a **value function** v
  Make sure that $v(D)$ = the cake, e.g., v_{c} or v_{m} for prediction decomposition
  ☞ **This is the tricky one!**

- Step 3: Choose an **efficient allocation**
  Summarizes the 2^d evaluations of v
  ☞ Switching the allocations can have **interesting properties**

Outline

1. Context and preliminaries

2. Cooperative games for feature importance

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Prediction decomposition - Baseline Shapley Values

Many methods have been proposed for **prediction decomposition** (Chen et al. 2023)

☞ Here $x \in \mathbb{R}^d$ is a data point for which the prediction is of interest

Baseline Shapley values:

$$v(A) = f(x_A, z_{\sim A}) \quad \Longrightarrow \quad \boxed{v(D) = f(x)}$$

where $z_{\sim A} \in \mathbb{R}^{|\sim A|}$ is some **baseline value** (e.g., $\mathbf{0}_{|\sim A|}$)

To evaluate $v(A)$ you need:

- An arbitrary baseline $z_{\sim A} \in \mathbb{R}^{|\sim A|}$
- The prediction on $f(x_A, z_{\sim A})$

Then, compute the **Shapley values**

Cost of redistributing $f(x)$: $\boxed{2^d \textbf{ model evaluations}}$

Prediction decomposition - Marginal Shapley values

Marginal Shapley values:

$$v(A) = \mathbb{E}\left[\widehat{f}(x_A, X_{\sim A})\right] \implies \boxed{v(D) = f(x)}$$

where the expectation is taken w.r.t. the law of the random vector $X_{\sim A}$

$v(A)$ is estimated using an i.i.d. sample $\mathbf{X}^n = (X^1, \ldots, X^n)$ with the same distribution as X

$$\widehat{v}(A, \mathbf{X}^n) = \frac{1}{n}\sum_{i=1}^{n} \widehat{f}\left(x_A, X^i_{\sim A}\right) \underset{n\to\infty}{\longrightarrow} \mathbb{E}\left[f\left(x_A, X_{\sim A}\right)\right] = v(A)$$

To compute an estimation $\widehat{v}(A, \mathbf{x}^n)$, you need:

- An i.i.d. dataset $\mathbf{x}^n$
- The predictions $f\left(x_A, x^i_{\sim A}\right)$ for $i = 1, \ldots, n$

Then, compute the (plug-in) **estimated marginal Shapley values**

Cost of redistributing $f(x)$: $\boxed{n \times 2^d \textbf{ model evaluations}}$

Prediction decomposition - Conditional Shapley values

Conditional Shapley values:

$$v(A) = \mathbb{E}\left[f(X) \mid X_A = x_A\right] \implies \boxed{v(D) = f(x)}$$

Monte Carlo estimation:

$v(A)$ is estimated using an i.i.d sample $\mathbf{X}^n_{\sim A|A} = (X^1_{\sim A|A}, \ldots, X^n_{\sim A|A})$ of $X_{\sim A} \mid X_A = x_A$

Sampling done using parametric assumptions, generative models...

$$\widehat{v}\left(A, \mathbf{X}^n_{\sim A|A}\right) = \frac{1}{n}\sum_{i=1}^{n} \widehat{f}\left(x_A, X^i_{\sim A|A}\right) \underset{n\to\infty}{\longrightarrow} \mathbb{E}\left[f(X) \mid X_A = x_A\right] = v(A)$$

To compute an estimation $\widehat{v}\left(A, \mathbf{x}^n_{\sim A|A}\right)$, you need:

- An observed sample $\mathbf{x}^n_{\sim A|A}$ drawn with cost c_{sam}
- The predictions $f\left(x_A, x^i_{\sim A|A}\right)$ for $i = 1, \ldots, n$

Then, compute the (plug-in) **estimated conditional Shapley values**

Cost of redistributing $f(x)$: $\boxed{n \times 2^d \textbf{ model evaluations} + nc_{\text{sam}} \times 2^d \textbf{ sampling cost}}$

Prediction decomposition - Conditional Shapley values

Conditional Shapley values:

$$v(A) = \mathbb{E}\left[f(X) \mid X_A = x_A\right] \implies \boxed{v(D) = f(x)}$$

Conditional expectation approximation:

Approximate $\mathbb{E}\left[f(X) \mid X_A\right]$ using an i.i.d. sample $\mathbf{X}^n$ of X and targets $\left(\widehat{f}\left(X^1\right), \ldots, \widehat{f}\left(X^n\right)\right)$

$$g_A^n \in \underset{\nu_A \in \mathcal{V}_A}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \ell\left(f\left(X^i\right), \nu_A\left(X_A^i\right)\right), \qquad \widehat{v}(A, \mathbf{X}^n) = g_A^n(x_A) \underbrace{\xrightarrow[n \to \infty]{}}_{\text{hopefully}} \mathbb{E}\left[f(X) \mid X_A = x_A\right] = v(A)$$

To compute an estimation $\widehat{v}(A, \mathbf{x}^n)$, you need:

- To fit g_A^n with cost c_{fit}
- Predict on $\mathbf{x}_n$ using g_A^n

Then, compute the (plug-in) **estimated conditional Shapley values**

Cost of redistributing $f(x)$: $\boxed{n\ \widehat{f}\ \textbf{evaluations} + n \times 2^d\ \widehat{g}\ \textbf{evaluations} + c_{\text{fit}} \times 2^d\ \textbf{learning cost}}$

Prediction decomposition - Other notable exemples

Interventional Shapley values:

Causal interpretation of the **marginal Shapley values**

☞ Under a flat causal directed acyclic graph (DAG) assumption

Asymmetric Shapley values: (Frye et al. 2020)

Leverages causal information from a DAG into the definition of the **allocation**

☞ Gives a bigger weight to parent variables

Banzhaf values: (Fumagalli et al. 2023)

Gives the same weight to all coalitions

☞ We loose the efficiency property

Owen value: (Miroshnikov et al. 2024) Same idea as the Banzhaf value

☞ But with the efficiency property

Variance decomposition - Shapley effects

Shapley effects: (Owen 2014)

$$v(A) = \frac{\mathbb{V}\left(\mathbb{E}\left[\widehat{f}(X) \mid X_A\right]\right)}{\mathbb{V}\left(\widehat{f}(X)\right)} \quad \Longrightarrow \quad \boxed{v(D) = 1}$$

Double Monte Carlo estimation :

To estimate $v(A)$:

1. Draw $\mathbf{X}_A^{n_v} = (X_A^1, \ldots, X_A^{n_v})$ i.i.d. with the same law as X_A
2. For each sample X_A^i, same estimator than the **conditional Shapley values**
3. **Empirical variance estimator** of the n_v estimations of the conditional expectation

☞ Compute the **estimated Shapley values**

Cost of redistributing $f(x)$: $\boxed{n_v \times \text{(cost of the conditional Shapley values)}}$

Variance decomposition - Proportional marginal effects

Proportional marginal effects: (Herin et al. 2024)

$$v(A) = \frac{\mathbb{E}\left[\mathbb{V}\left(\widehat{f}(X) \mid X_A\right)\right]}{\mathbb{V}\left(\widehat{f}(X)\right)} \implies \boxed{v(D) = 1}$$

Double Monte Carlo estimation :

To estimate $v(A)$:

1. Draw $\mathbf{X}_A^{n_v} = (X_A^1, \ldots, X_A^{n_v})$ i.i.d. with the same law as X_A
2. For each sample X_A^i, same estimator than the **conditional Shapley values**
3. **Empirical variance estimator** of the n_v estimations of the conditional expectation

☞ Compute the **estimated proportional values**

Cost of redistributing $f(x)$: $\boxed{n_v \times \text{(cost of the conditional Shapley values)}}$

Variance decomposition - Interpretation

Variance decompositions: interpreted as attributed **percentages of the model's uncertainty**

This comes from the fact that the **value function** is **monotonic**

$$\forall A \in \mathcal{P}_D,\ \forall B \subsetneq A, \quad v(B) \leq v(A)$$

☞ For every coalitions, their sub-coalition have a smaller (or equal) value

In this case, the **allocation** is **positive**

$$\forall \pi \in \mathcal{S}_D,\ \forall i \in D \quad \mathrm{MC}^i_\pi = v(\pi^i) - v(\pi^i \setminus \{j\}) \geq 0 \quad \Longrightarrow \quad \phi(i) = \sum_{\pi \in \mathcal{S}_D} \underbrace{p(\pi)}_{\in [0,1]} \times \underbrace{\mathrm{MC}^i_\pi}_{\geq 0} \geq 0$$

Holds for $\mathbb{V}\left(\mathbb{E}\left[\widehat{f}(X) \mid X_A\right]\right)$ because of the **contractive** property of conditional expectations

$$\forall A \in \mathcal{P}_D,\ \forall B \subsetneq A, \quad \mathbb{E}\left[\left(\mathbb{E}\left[\widehat{f}(X) \mid X_B\right]\right)^2\right] \leq \mathbb{E}\left[\left(\mathbb{E}\left[\widehat{f}(X) \mid X_A\right]\right)^2\right]$$

☞ Same thing as the R^2 decreasing with the number of covariates in linear regression

Exotic decompositions

Many quantities of interest (i.e., cakes) have been studied in the literature

☞ **Uncertainty measures in reproducing kernel Hilbert spaces** (da Veiga 2021)

☞ **Uncertainty measures based on entropy** (Watson et al. 2023)

☞ **Algorithmic fairness measures** (Bénesse et al. 2024)

☞ **Uncertainty measures based on conformal prediction intervals** (I. et al. 2025c)

☞ **Indirect discrimination via proxies** (Lindholm et al. 2026)

Reducing the computational burden

These methods are **computationally expensive**

The complexity is proportional to 2^d

☞ Grows exponentially fast with the number of covariates

Due to the fact that we compute v for **every** coalition

Two strategies to break the exponential cost:

1. Leverage **extra knowledge** on the structure of $\widehat{f}$ (e.g., linear model, random forest)
   ☞ Defeats the "model-agnostic" purpose
2. **Compute v only on a selected set of coalitions**

Sampling coalitions - Monte Carlo and random orders

A simple idea, introduced by Štrumbelj et al. (2014), leverages **random orders**

Recall that we aim to estimate

$$\phi(i) := \mathbb{E}_{\pi \sim p}\left[v\left(\{\pi_1, \ldots, \pi_{\pi(i)}\}\right) - v\left(\{\pi_1, \ldots, \pi_{\pi(i)-1}\}\right)\right]$$

With an i.i.d. sample $\pi^1, \ldots, \pi^m$ drawn according to the pmf p the estimator

$$\widehat{\phi}(i) = \frac{1}{m}\sum_{j=1}^{m}\left[v\left(\left\{\pi_1^j, \ldots, \pi_{\pi(i)}^j\right\}\right) - v\left(\left\{\pi_1^j, \ldots, \pi_{\pi(i)-1}^j\right\}\right)\right]$$

is an **unbiased**, **strongly consistent** and **asymptotically normal** estimator of $\phi(i)$

Each sampled order π^i selects **<u>at most</u>** d coalitions

☞ **Only evaluate v for $m \times d$ coalitions instead of 2^d (worst-case)**

Sampling coalitions - Importance sampling and random orders

Let's say you have p and p' leading to allocations ϕ and ϕ'

You **already sampled** from p, and computed $\widehat{\phi}$ on a select subset of coalitions

You can **recycle** the estimates of v to compute

$$\widehat{\phi'}(i) = \frac{1}{m} \sum_{i=1}^{m} \frac{p'(\pi_i)}{p(\pi_i)} \left[v\left(\{\pi_1, \ldots, \pi_{\pi(i)}\}\right) - v\left(\{\pi_1, \ldots, \pi_{\pi(i)-1}\}\right) \right]$$

is an **unbiased**, **strongly consistent** and **asymptotically normal** estimator of $\phi'(i)$

☞ **Two allocations for the cost of one**

Outline

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4. Discussion

Discussion

Allocations are a "**mechanical decomposition**" of virtually any quantity of interest
☞ Thanks to **telescopic sum properties**

We saw that there can be **several value functions** for the **same quantity of interest**
e.g., **conditional expectation** and **marginal expectation** for prediction decomposition

Which one is the the "right" one?

Several ways to **assess the validity of a decomposition**:

1. **Human-centered evaluations**
   ☞ Participants evaluate usefulness, trust, transparency...

2. **Towards a formalized "canonical decomposition"**
   Math-heavy, with promising breakthroughs, but still very much an open field

Canonical decomposition - Independent covariates

Recall that the **Harsanyi dividends** using **conditional expectation** writes

$$\mathcal{D}_{v_c}(A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} \mathbb{E}\left[\widehat{f}(X) \mid X_B\right]$$

If X were to be **mutually independent**, Hoeffding (1948) showed there is a unique way to write $\widehat{f}(X)$

$$\widehat{f}(X) = f_\emptyset + \sum_{i=1}^{d} f_i(X_i) + \sum_{i \neq j} f_{ij}(X_i, X_j) + \cdots + f_{1,\ldots,d}(X_1, \ldots, X_d) = \sum_{A \in \mathcal{P}_D} f_A(X_A)$$

such that **all summands are pairwise orthogonal**

In that case, he showed that

$$f_A(X_A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} \mathbb{E}\left[\widehat{f}(X) \mid X_B\right]$$

☞ The **conditional Shapley values** have a "theoretical meaning" **when the covariates are mutually independent**

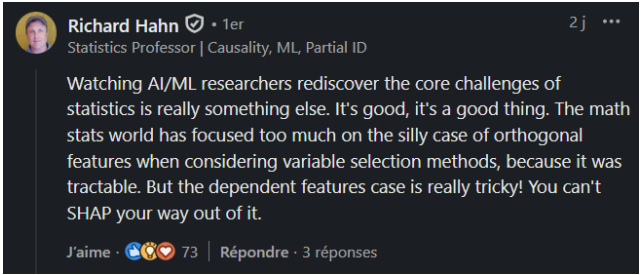
Canonical decomposition - Dependent covariates

If the covariates are **not independent**, it is **still possible to define a canonical decomposition**

☞ Under a **hierarchical orthogonality** constraint (Hooker 2007; Chastaing et al. 2012; I. et al. 2025a)

No straightforward **model-agnostic estimation schemes**, but some are **model-specific** :

- When $\widehat{f}$ is a **boosting model** (Champion et al. 2015)
- When $\widehat{f}$ is a **random forest** (Benard 2025)
- When the **covariates are Bernoulli** (Ferrere et al. 2025)



☞ **This is very much still an open-problem**

Conclusion

What we covered:

- Cooperative games offer a way to attribute importance to covariates
  ☞ Three ingredients: a **quantity of interest**, a **value function**, and an **efficient allocation**
- The **Harsanyi set** and the **Weber set** **families of allocations**
  ☞ The **Shapley values** belong to both and the **Weber set** is a strict subset of the **Harsanyi set**
- **Prediction decomposition** attributes $\widehat{f}(x)$ to each covariate
  ☞ Many choices of **value function** e.g., **marginal** vs. **conditional** expectation
- **Variance decomposition** attributes $\mathbb{V}\left(\widehat{f}(X)\right)$ to each covariate
  ☞ Shapley effects, proportional marginal effects (PME), and beyond
- The **exponential cost** in 2^d can be mitigated via sampling strategies

The big open question:

- No consensus on a meaningful **value function** for dependent covariates
- **Better vs. different debate** when it comes to decompositions

Cooperative games are great tools, but we're still not sure on how to use them

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Thank you for your attention!

Any questions?

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